JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 88, NO. B4, PAGES 3523-3529, APRIL 10, 1983

PALEOMAGNETISM OF MESOZOIC AND TERTIARY ROCKS FROM EAST EL OWENAT AREA, SOUTHWEST, EGYPT

A. Gouda Hussain and Y. Aziz

Helwan Institute of Astronomy and Geophysics, Helwan, Cairo, Egypt

Abstract. Ten sites (310 samples) from alkali syenite at East El Oweinat (23.2°N, 29.5°E) yield a stable mean direction of magnetization D = 340.3, I = 19.9 with α_{95} = 12.1, K = 13.4, N = 10 after alternating field cleaning resulting in a pole at 67.8°N. 268.8°E with A_{95} = 8.6°. This pole position agrees with the other Upper Cretaceous pole positions of Africa. From seven sites (82 samples) from Upper Cretaceous Nubian sandstone at East El Oweinat a mean direction D = 349.9, I = 27.1 with α_{95} = 10.1, K = 37.8 (mixed polarity) was obtained after thermal demagnetization. Combined with previous investigations on Nubian sandstone at other localities (Schult et. al., 1978, 1981), this yields a pole at 80.8°N, 238.9°E with A_{95} = 3.2, K = 435.9, N = 30. Five sites from Tertiary basalt yield a mean direction D = 12.4, I = 23.9 and α_{95} = 9.2, K = 46.6 after alternating field cleaning which yield a paleopole position at 74.1°N, 159.8°E with A_{95} = 6.0, K = 107.8, N = 5. The paleopole positions of East El Oweinant rocks confirm the apparent polar wander path of Africa established in previous papers (Hussain et. al., 1980; Schult et. al., 1981). More comparisons with rotated South American and Australian pole positions after proper reconstructions to Africa show agreements with the present results.

1. Introduction

In some previous papers [Hussain et al., 1979; Schult et al., 1978, 1981; Hussain, 1977] the paleomagnetism of Mesozoic and Tertiary rocks in Egypt was studied. The polar wander path of Africa during the Mesozoic and Tertiary was produced [Hargraves and Onstott, 1980; Hussain et al., 1980] and improved in some detailed fashion by Schult et al. [1981]. More improvements and correlations with the pole positions in Mesozoic and Tertiary is desirable. In this paper the paleomagnetism of Mesozoic alkali syenite from Tertiary basalt as well as Upper Cretaceous Nubian sandstone from East El Oweinat, SW of Egypt, is presented. The paleopole positions from these rock units may add informations to the pole wander path of Africa.

2. Geological Setting

Figure 1 shows a geological sketch map of East El Oweinat area and the sampling sites. The geological information about this area is rather limited. A detailed surface survey for the southern part of the Western Desert between the Nile and longitude 29°E was first done by Issawi [1971]. El Ramli [1972] made more surveys of El Oweinat area and published a compiled geological map for the southern western part of Egypt. Figure 1 is a part of this compiled map [El Ramli, 1972]. Issawi and El Ramli, according to stratigraphic reasons, detected a Mesozoic age for the syenites and a Tertiary age for the basalt. Unfortunately, no radiometric age determinations for these volcanic rocks have been made. On the other hand, the Nubian sandstone extends over large parts of southern Egypt and previously has been paleomagnetically studied at other localities in Egypt [El Shazly and Krs, 1973; Hussain, 1977; Schult et al., 1978, 1981]. The age of the Nubian sandstone is Upper Cretaceous [Said, 1971].

Samples were collected from 10 sites from alkali syenite, five sites from Tertiary basalts and from seven sites from Nubian sandstone (Figure 1) where no tectonic movements are recognizable.

3. Sampling and Measurements

About six oriented blocks were collected from each site. Cores of 2.5 cm in diameter were drilled from blocks and cut into specimens of 2.5 cm length. The natural remanent magnetization (NRM) was measured using a spinner flux gate magnetometer. Three pilot samples from each site of the volcanics were then subjected to alternating field (af) demagnetization in steps up to 1000 Oe (10^4 Oe = 1 Tesla) peak value. In Figures 2c-2e vector diagrams show typical examples for the removal of viscous magnetization by af treatment. All other samples were subjected to af demagnetization with appropriate peak values according to the vector diagrams for the pilot samples, and the stable remanent magnetization since af treatment had no effect on direction or intensity of remanent magnetization. Typical vector diagrams of thermal demagnetization of pilot samples are shown in Figures 2a and 2b and Figures 3a-3c from which the appropriate demagnetization temperature for all samples was chosen.

4. Results

4.1. Mesozoic Alkali Syenite

The results are summarized in Table 1 and the site mean directions with their circles of confidence of the stable remanent magnetization (characteristic remanent magnetization (CARM)) are shown in Figure 4. The within site scatter as well as the between site scatter were reduced by af treatment. All sites have normal directions of magnetization and all of them have been considered to be consistent. Giving unit weight to each site, the mean direction of the CARM of 10 sites is D = 340.3°, I = 19.9°, with α_{95} = 12.1°. The paleopole positions due to each site and the mean of all paleopoles are represented in Figure 5. This mean value (268.8°E, 67.8°N, with A_{95} =

Copyright 1983 by the American Geophysical Union.

Paper number 2B0894.
0148-0227/83/002B-0894$05.00

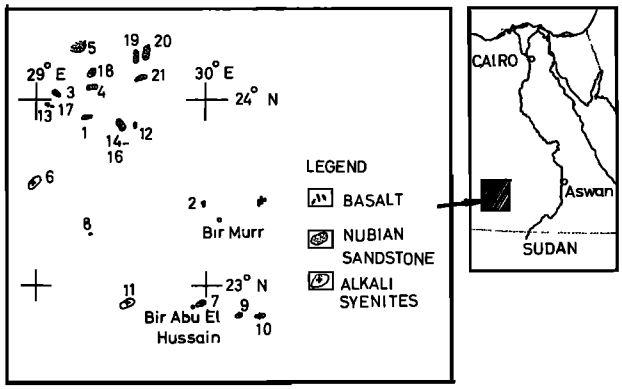


Fig. 1. Geological sketch map of East El Oweinat Area [El Ramli, 1972] showing sampling sites. Numbers refers to site. Numbers refer to site numbers.

8.6) is in agreement with other Mesozoic paleopole positions of Africa [Schult et. al., 1981].

4.2. East El Oweinat Nubian Sandstone

The results are listed in Table 2 and shown in Figures 6 and 7. Six sites from seven have normal polarity, and only one site has reversed polarity. Secondary soft magnetization with reversed polarity has been observed during thermal demagnetization (Figure 3), which has been cleaned by low temperatures (100°-300°C). This secondary magnetization could be due to hydrothermal solutions of age not too much younger than the deposition of the Nubian sandstone.

The mean direction of all site means is D = 349.9°, I = 27.1° with α_{95} = 10.1. The paleopole positions and their mean value together with their circle of confidence are represented in Figure 7. This mean value (258.2°E, 76.7°N with A_{95} = 8.6) is rather different from the pole position from the Wadi Natash volcanics and is in good agreement with the pole positions from Nubian sandstone from Wadi Natash area, Qena-Safaga, Idfu-Mersa Alam and Aswan areas reached in previous papers [Schult et al., 1978, 1981]. It may be better to get a compiled paleopole position for the Nubian sandstone, giving unit weight to each site. This yields a pole position at 238.9°E, 80.8°N with A_{95} = 3.2 (mean of this paper plus Schult et al. [1978, 1981]; 16 normal polarity and 14 reversed) which is slightly different from the present dipole position (Figure 9). The mixed polarity of the Nubian sandstone may be correlated with the age of 78 m.y. B.P. at which the geomagnetic field reversed its direction after long normal polarity period [Lowrie et al., 1980].

4.3. Tertiary Basalt

The paleomagnetic results from the Tertiary basalt are summarized in Table 3, and the mean site directions together with their circles of confidence of the stable magnetization are represented in Figure 8a. All sites have normal polarity which is not typical for Tertiary basalts from Egypt [Hussain et al., 1976, 1979].

The mean direction of all site means is D = 12.4°, I = 23.9° with α_{95} = 9.2°. The paleopole positions due to each site and the mean of all are represented in Figure 8b (159.8°E, 74.1°N, A_{95} = 6.0) which agrees with other African Tertiary pole positions [Hussain, 1977; Hussain et al., 1976, 1979; Schult and Soffel, 1973].

5. Discussion

The paleopole position from Mesozoic syenite from East El Oweinat agrees with African Upper Cretaceous poles (Figure 9a). This pole position and the positive magnetic polarity of all sites may lead to an Upper Cretaceous age for the syenite earlier than 78 m.y., the age after which the geomagnetic field polarity was reversed after

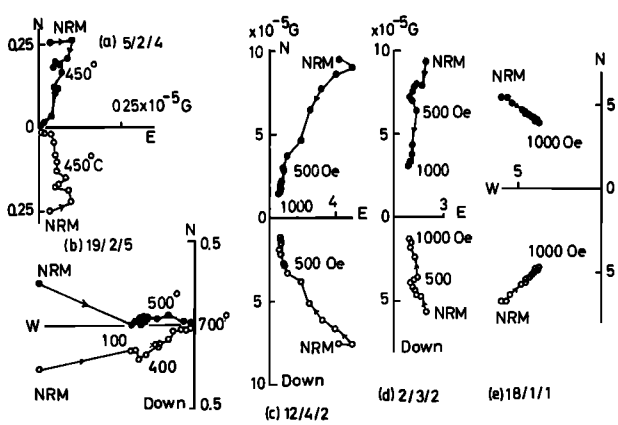


Fig. 2. Figures 2c-2e are vector diagrams showing the variation of the remanence vector during progressive af demagnetization of samples from Mesozoic and Tertiary volcanic rocks. Open and solid circles indicate components in the vertical EW and in the horizontal planes, respectively. The remanence is relatively stable specially in Figure 2e. Figures 2a and 2b are as in Figure 3. 1 G = 10^3 A m^{-1}, 1 Oe = 10^{-4} T.

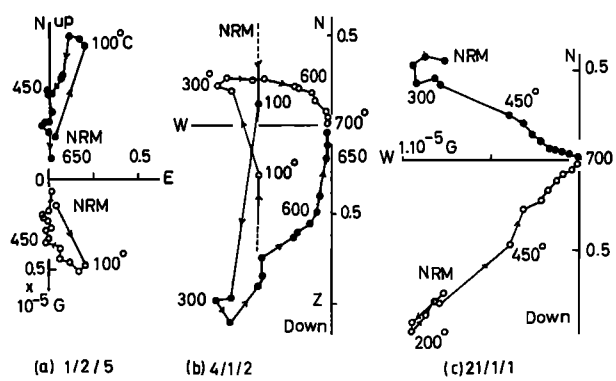


Fig. 3. Vector diagram showing the variation of the remanence vectors during progressive thermal demagnetization of samples from Nubian sandstone. Open and solid circles indicate components in the vertical EW and the horizontal planes, respectively. These examples show overprinting by reversed instable components in Figures 3a and 3c, where the CARM components are normal, and overprinting of normal instable components in Figure 3b where the CARM components are reversed.

21562202b, 1983, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB088iB04p03523 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Site Mean Paleomagnetic Results of Mesozoic East El Oweinat Volcanics

		NRM					CARM				VGP	
Site	N	D, deg	I, deg	α_{95}	k	AF	N	D, deg	I, deg	α_{95}	k	^{o}E	^{o}N
6	65	348.6	20.7	14.2	10.8	500	61	343.0	8.7	8.5	10.6	242.3	57.5
7	6	355.1	9.0	29.3	3.9	500	6	333.4	2.6	18.2	4.4	262.3	56.0
9	55	343.1	41.1	19.7	6.3	600	52	334.2	56.9	11.0	18.5	336.6	64.2
10	30	357.1	42.4	5.9	79.3	600	30	353.9	41.7	5.9	26.6	300.2	84.5
11	66	349.3	31.7	10.8	12.9	500	66	356.0	33.8	6.6	15.8	250.4	84.1
12 b	10	336.8	8.6	12.5	7.2	500	10	336.4	11.1	9.1	16.4	263.7	60.8
14	23	329.2	9.5	16.5	11.8	500	23	329.3	5.5	8.8	13.9	267.6	53.9
15	21	340.1	37.1	5.5	11.7	500	21	334.4	28.0	7.5	47.7	283.0	64.2
16	20	353.5	19.9	7.2	25.2	550	17	351.7	3.4	10.0	42.0	230.0	66.3
18	15	322.6	42.5	9.3	25.9	550	14	333.1	23.3	8.9	20.4	278.9	61.9
Mean of site means							10	340.3	19.9	12.1	13.4	268.8	67.8

A_{95} = 8.6, K = 26.4

N, number of samples; D, declination; I, inclination, α_{95} or A_{95}, radius of 95% confidence circle; k or K, precision parameter; AF, peak value of alternating field used for demagnetization (10^4 Oe = 1 T).

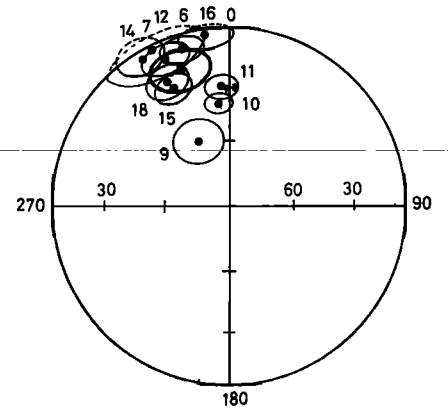


Fig. 4. Equal-area projection of site mean directions with their circles of confidence for Mesozoic rocks from East El Oweinat. Square with thick ellipse denotes the overall mean. Solid symbols indicate dipole field direction and earth's field direction, respectively.

a long positive polarity sequence [Lowrie et al., 1980].

On the other hand, the pole position from Upper Cretaceous Nubian sandstone from East El Oweinat nearly coincides with other poles from Nubian sandstone from other localities [Hussain, 1977; Schult et al. 1978, 1981]. This pole position and the mixed polarity of the magnetization of the Nubian sandstone may lead to an age younger than the Upper Cretaceous volcanics from the Eastern Desert [Schult et al., 1981; Ressetar et al., 1981]. This nearly unique direction of magnetization of the Nubian sandstone from different localities (also from different strata at Aswan [Hussain et al., 1976] is rather unusual for such thick, porous, and widely spreading sediments. Much more studies about the magnetization and magnetostratigraphy of the Nubian sandstone at different localities are needed before making any conclusion about the age of magnetization of these sediments and whether it is later than its deposition or not.

An attempt to define the apparent pole wander-path (APWP) for Mesozoic to Tertiary for Africa is

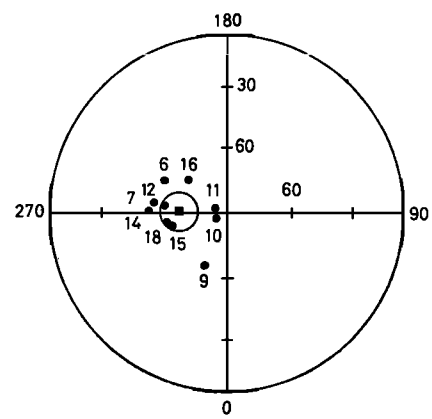


Fig. 5. Equal-area projection of paleopole positions for each site from Mesozoic volcanics. Square with circle of confidence represents the mean paleopole position.

21562202b, 1983, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB088iB04p03523 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Site Mean Paleomagnetic Results for Nubian Sandstone From East El Oweinat (ca. 23.6°N, 28.6°E)

		NRM					CARM				VGP	
Site	N	D, deg	I, deg	α_{95}	k	T, °C	N	D, deg	I, deg	α_{95}	k	°E	°N
1	15	355.3	25.9	6.2	17.6	500	15	351.8	22.2	4.7	66.6	242.4	75.3
3	15	358.1	32.1	3.8	138.9	450	15	352.0	13.2	6.3	146.8	234.2	74.3
4	12	355.9	32.5	5.5	18.7	500	11	186.2	-18.5	5.4	72.4	185.8	74.3
5	10	354.2	47.8	5.6	18.2	430	10	350.8	50.7	5.9	18.3	343.8	79.0
19	12	356.8	27.5	6.4	18.3	400	11	353.8	22.6	7.2	146.7	235.8	76.4
20	10	340.0	27.7	6.8	146.7	540	9	339.6	27.7	6.3	115.8	276.9	68.7
21	11	331.6	31.6	6.6	29.9	500	11	332.6	33.1	5.9	59.7	291.2	63.8
Mean of site means*							7	349.9	27.1	10.1	37.8	258.2	76.7
										A_{95} = 8.6			
Mean of site means**							30					238.9	80.8
										A_{95} = 3.2, K = 435.9			

See Table 1 footnotes.
*Six normal, one reversed.
**This paper plus Schult et al. [1978, 1981]; 16 normal, 14 reversed.

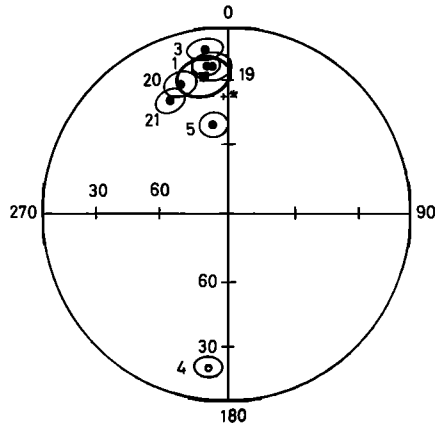


Fig. 6. The same as Figure 4, for the Upper Cretaceous Nubian sandstone from East El Oweinat.

represented in Figure 9a. Such attempts were done before by Martin et al. [1978], Hargraves and Onstott [1980], Hussain et al. [1980], and Schult et al. [1981]. Although the present results and some other recent results [Ressetar et al., 1981] add more information to the African Mesozoic poles, no significant modification to our later APWP [Schult et al., 1981] could be made, since these latest results nearly coincide with the earlier results of Wadi Natash volcanics.

The Mesozoic African poles are listed in Table 4 except some other results [i.e., El Shazly and Krs, 1973; McFadden and Jones, 1977] which were modified later. The results of El Shazly and Krs [1973] for Wadi Natash volcanics show relatively large scatter with k = 4.85 and would yield a larger circle of confidence about the mean, giving unit weight to each site rather than to each sample, as was done by El Shazly and Krs [1973]. This Upper Cretaceous Wadi Natash pole was improved by Schult et al. [1981] and confirmed by Ressetar et al. [1981]. The later Upper Cretaceous pole from kimberlite pipes [McFadden and Jones, 1977] was recently improved by Hargraves and Onstott [1980].

The rotated South American poles (Table 4, Figure 9b) show good agreement and possess the same loop as the African poles. This good coin-

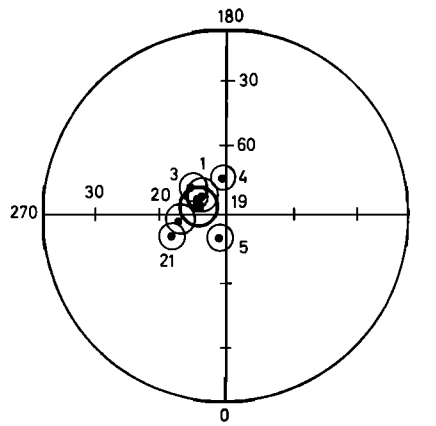


Fig. 7. The same as Figure 5, for the Nubian sandstone from East El Oweinat.

21562302b, 1983, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB088iB04p03523 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3. Site Mean Paleomagnetic Results of East El Oweinat Tertiary Basalt (Mean Location, 23.3°N, 29.0°E)

		NRM					CARM				VGP	
Site	N	D, deg	I, deg	α_{95}	k	AF, Oe	N	D, deg	I, deg	α_{95}	k	°E	°N
2	8	5.0	30.2	4.4	330.9	500	8	4.9	28.8	4.4	332.0	177.0	80.9
8	100	8.6	34.9	8.6	17.7	550	87	7.3	26.8	6.2	27.5	171.2	78.4
12	77	15.5	3.9	7.2	32.7	550	72	17.1	4.8	7.9	92.0	167.8	63.6
13	43	18.2	28.0	5.9	18.9	550	42	17.1	27.7	5.9	29.1	145.9	71.4
17	17	15.4	31.5	7.0	20.4	500	11	14.8	30.7	7.2	30.5	144.5	74.2
Mean of site means*							5	12.4	23.9	9.2	46.6	159.8	74.1
												A95 = 6.0	
												K = 107.8	

See Table 1 footnotes.
*All positive.

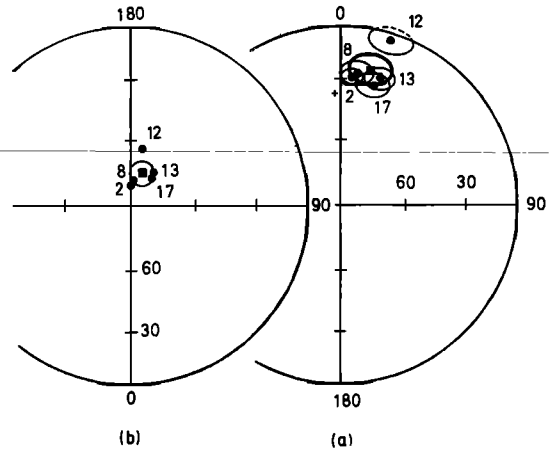


Fig. 8. (a) The same as Figure 4. The same as Figure 5 for the Tertiary basalt from East El Oweinat.

cidence between the African and South American poles could be achieved by assuming a small separation in the southern Atlantic in early and middle Mesozoic time [Schult et al., 1981].

Moreover, recent results from Australia [Schmidt and Embleton, 1981] show a loop during Mesozoic as the African and South American loops. The rotated poles of Australia for reconstructions based on the work by Norton and Sclater [1979] are listed in Table 4 and represented in Figure 9c with the same labels as Figure 9a for nearly the same ages. The Australian rotated poles labeled 1 to 4, 9 and 10 (Figure 9c) are in good agreement with the African and South American poles of nearly the same ages. Poles 5 to 7 describe the loop during Jurassic and Lower Cretaceous. In case of African poles, pole labeled 5 was derived from Morroccan volcanics which are rocks located

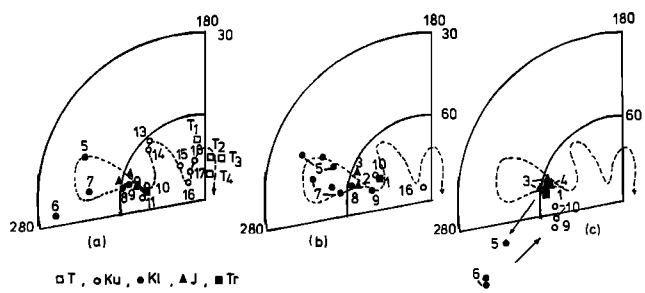


Fig. 9. (a) Mesozoic and some Tertiary pole positions from Africa and proposed apparent polar wander path. The dotted line indicates the trend of the pole positions and the path is of variable width depending on α_{95} values of the poles. Labels 1-18 refer to Table 4. Tertiary poles T_1 upper Eocene rocks in northern Egypt [Hussain et al., 1979], T_2 Tertiary basalt of Ethiopia [Schult, 1974], T_3 East El Oweinat basalt (Table 3), T_4 Eocene Bahariya Fe Ores [Schult et al., 1978]. (b) Rotated South American poles [Schult et al., 1981]. (c) Rotated Australian poles, rotation parameters for Australia, see Table 4. Same labels in Figures 9a-9c refer to poles with approximately the same ages. The indicated polar wander path is reproduced from Figure 9a.

21562202b, 1983, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB088iB04p03523 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4. Mesozoic Paleomagnetic Poles for Africa, South America, and Australia

	Africa					South America				Australia			
		Age,	Pole Position			Age,	Pole Position			Age,	Pole Position		
Pole	Formation	m.y.	°N	°E	A95	m.y.	°N	°E	A95	m.y.	°N	°E	A95
1	Trias mean		69	263	4.9		70	274	6.3	185-193	58	270	10
2	Stormberg, Karoo	154-190	65	262	11.7	157-173	63	261	6	170	59	260	6
3	Hoachanas	161-173	62	262	7	158±12	62	250	6.9	170	62	287	-
4	Mateka Hills	168	59	260	8.3					<Tr	64	257	4
										<Tr	62	250	8
										<Tr	63	257	5
5	Morroccan volcanics	Kl	44	251	5	129-150	41	251	6				
						123±4	53	252	10	<J	45	290	7
						121±3	48	249	8				
6	Kimberlite pipes	122-162	36	277	17					150	34	299	16
										133	33	304	4
7	Kaoko lavas	110-128	48	267	2	115-130	47	261	5.7				
						119±5	57	266	3.7				
						118±6	54	264	1.9				
8	Mlanje Massif	116-128	60	262	12	121	60	261	9				
9	Lupata volcanics	106-111	62	260	3.5	110	68	261	4.5	>100	69	285	10
10	Wadi Nastash volcanics	77-100	69	258	8.5	85-99	68	247	4.5	100	69	273	11
										100	68	262	10
11	East El Oweinat	Ku	68	269	8.6								
12	W El Qusseir Trachytes + ring complex	63-92	65	249	11.4								
13	Kimberlite pipes	82-88	61	224	7.8								
14	Sicily volcanics	71-81	62	229	3.8								
15	Morroccan sediments	Kl-Ku	75	217	5.5								
16	Nubian sandstone	Ku	77	258	3.2	63-80	85	211	10				
17	Red siltstone	Ku	79	208	6								
18	Tororo	Ku(?)	76	195	9								

African poles: 12, Ressetar et al. [1981]; 11 and 16, this paper; 6 and 13, Hargraves and Onstott [1980]; other poles see Hussain et al. [1980] and Schult et al. [1981]. South American rotated poles see Schult et al. [1981]. Australian rotated poles [Schmidt and Embleton, 1981] and rotation parameters (based on Norton and Sclater [1979]) are -50.5° about a pole at 24.6°N, 119.6°E for 130 m.y. and -33.2° about a pole at 22.8°N, 97°E for 100 m.y. (P.W. Schmidt and B.J.J. Embleton, personal communication, 1981).

21562202b, 1983, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB088iB04p03523 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

in the tectonically active Atlas area [Martin et al., 1978] while pole 6 is from kimberlite pipes [Hargraves and Onstott, 1980] and was derived from only four sites with a relatively large circle of confidence. The corresponding Australian rotated poles (labeled 5, Figure 9c) are rather different in position, while the South American poles (labeled 5 and 7, Figure 9b) are in good agreement with the corresponding African ones. The APWP for the three continents possess a loop during Mesozoic; much more results from rocks from Jurassic and Lower Cretaceous, although they are needed to describe the exact shape of this loop.

On the other hand, the Tertiary paleopoles from Africa are rather scattered [Hussain et al., 1976, 1979; Hussain, 1977; Schult, 1974; Schult and Soffel, 1973; Schult et al., 1981; Ressetar et al., 1981]. The pole from East El Oweinat Tertiary basalt agrees with most of them. The APWP during Tertiary can be tentatively drawn as in Figure 9a. Actually, more data are also needed for further adjustment of the APWP of Africa in Tertiary.

Acknowledgments. We thank M. Fahim, H. Deebes, and A. Gh. Hassanain, Helwan Institute of Astronomy and Geophysics, Cairo; G. Angenheister, H. Soffel, and A. Schult, Institut fur Allgemeine und Angewandte Geophysik, Universitat Munchen for their help and support. We also thank Hussain Kamel, General Petrolium Company of Egypt (GPC) for providing the opportunity to collect samples from this unaccessible area during some prospecting work of the GPC there.

References

El Ramli, M. F. A new geological map for the basement rocks in the Eastern and South-Western deserts of Egypt, scale 1:1,000,000, Ann. Geol. Surv. Egypt, 2, 1-18, 1972.

El Shazly, E. M., and M. Krs, Palaeography and palaeomagnetism of the Nubian sandstone, Eastern Desert of Egypt, Geol. Rundsch,. 62, 212-225, 1973.

Hargraves, R. B., and T. C. Onstott, Palaeomagnetic results from some southern African kimberlites and their tectonic significance, J. Geophys. Res., 85(B7), 3587-3596, 1980.

Hussain, A. G., Magnetization and palaeomagnetism of some sedimentary rock units in Bahariya area, Western Desert, Egypt, Helwan Observ. Bull., 141, 1-20, 1977.

Hussain, A. G., A. Schult, H. Soffel, and M. Fahim, Magnetization and palaeomagnetism of Abu Zaabal and Abu Rawash basalts, Egypt, Helwan Observ. Bull., 134, 1-15, 1976.

Hussain, A. G., A. Schult, and H. Soffel, Palaeomagnetism of basalt of Wadi Abu Tereifiya, Mandisha and dioritic dykes of Wadi Abu Shihat, Egypt, Geophys. J. R. Astron. Soc., 56, 55-61, 1979.

Hussain, A. G., A. Schult, and H. Soffel, Preliminary polar wander path of Egypt between Cretaceous and Quaternary, Berl. Geowiss. Abh. Reihe A, 19, 93-94, 1980.

Issawi, B., Geology of Derb El Arbain Area, Ann. Geol. Surv. Egypt, I, 53-92, 1971.

Lowrie, W., J. E. T. Channell, and W. Alvarez, A review of magnetic stratigraphy investigations in Cretaceous pelagic carbonate rocks, J. Geophys. Res., 85(B7), 3597-3605, 1980.

Martin, D. L., A. E. M. Nairn, H. C. Noltimier, M. H. Petty, and T. J. Schmitt, Palaeozoic and Mesozoic palaeomagnetic results from Morrocco, Tectonophysics, 44, 91-114, 1978.

McFadden, P. L., and D. L. Jones, The palaeomagnetism of some upper Cretaceous Kimberlite occurrences in South Africa, Earth Planet. Sci. Lett., 34, 125-135, 1977.

Norton, I. O., and J. G. Sclater, A model for evolution of the Indian Ocean and the breakup of Gondwanaland, J. Geophys. Res., 84(B12), 6803-6829, 1979.

Ressetar, R., A. E. M. Nairn, and J. R. Monrad, Two phases of Cretaceous-Tertiary magnetism in the Eastern Desert of Egypt: Palaeomagnetic, chemical and K-Ar evidence, Tectonophysics, 73, 169-193, 1981.

Said, R., Explanatory notes to accompany the geological map of Egypt, Pap. Geol. Surv. Egypt, 56, 1971.

Schmidt, P.W., and B. J. J. Embleton, Magnetic overprinting in southeastern Australia and the thermal history of its rifted margin, J. Geophys. Res., 86(B5), 3998-4008, 1981.

Schult, A., Palaeomagnetism of Tertiary volcanic rocks from the Ethiopian Southern and Danakil Block, J. Geophys., 40, 203-212, 1974.

Schult, A., and H. Soffel, Palaeomagnetism of Tertiary basalts from Libya, Geophys. J. R. Astron. Soc., 32, 373-380, 1973.

Schult, A., H. Soffel and A. G. Hussain, Palaeomagnetism of Cretaceous Nubian sandstone, Egypt, J. Geophys., 44, 333-340, 1978.

Schult, A., A. G. Hussain, and H.C. Soffel, Palaeomagnetism of Upper Cretaceous volcanics and Nubian sandstone of Wadi Natash, SE Egypt and implications on the polar wander path for Africa in the Mesozoic, J. Geophys., 50, 16-22, 1981.

(Received November 17, 1981;
revised June 1, 1982;
accepted June 11, 1982.)

21562202b, 1983, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB088iB04p03523 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License